HWA CHONG INSTITUTION
Preliminary Examination
Higher 1

CANDIDATE NAME		**CT GROUP**	11S
CENTRE NUMBER		**INDEX NUMBER**	

CHEMISTRY **8872/01**

Paper 1 Multiple Choice **21 September 2012**

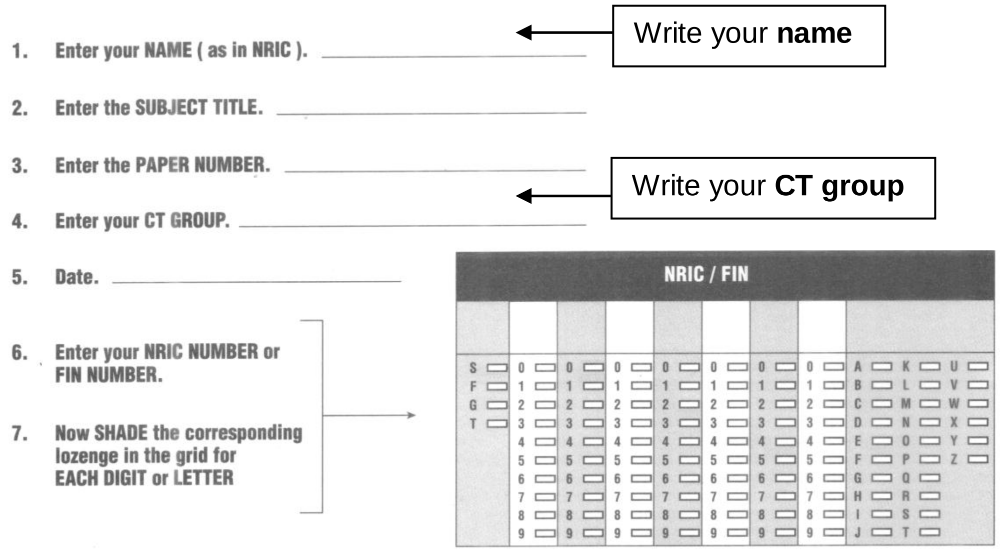
50 min

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Complete the information on the optical mark sheet (OMS) as shown below.



Write your **name**

Write your **CT group**

Write <u>and</u> shade your **NRIC** or **FIN number**

There are **thirty** questions on this paper. Answer **all** questions. For each question, there are four possible answers **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the OMS.

Each correct answer will score one mark. A mark will **not** be deducted for a wrong answer.

Any rough working should be done in this booklet.

Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the one you consider to be correct.

1 *Use of the Data Booklet is relevant to this question*.

Analytical chemists can detect very small amounts of amino acids, down to 3×10^{-21} mol.

How many molecules of an amino acid would this be?

A 9 **B** 200 **C** 1 800 **D** 360 000

2 Carbon disulfide vapour burns in oxygen according to the following equation.

$$CS_2(g) + 3O_2(g) \rightarrow CO_2(g) + 2SO_2(g)$$

A sample of 10 cm^3 of carbon disulfide was burned in 50 cm^3 of oxygen. After measuring the volume of gas remaining, the product was treated with an excess of aqueous sodium hydroxide and the volume of gas measured again. All measurements were made at the same temperature and pressure; under such conditions that carbon disulfide was gaseous.

What were the measured volumes?

	Volume of gas after burning/ cm^3	Volume of gas after adding NaOH(aq)/ cm^3
A	30	0
B	30	20
C	50	20
D	50	50

3 Which statement explains why the boiling point of methane is higher than that of neon? [Ar: H, 1; C, 12; Ne, 20]

A A molecule of methane has a greater mass than a molecule of neon.

B Molecules of methane have stronger intermolecular forces than those of neon.

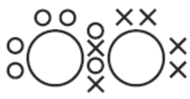
C Molecules of methane form hydrogen bonds, but those of neon do not.

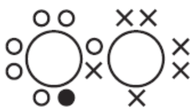
D The molecule of methane is polar, but that of neon is not.

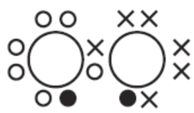
4 When barium metal burns in oxygen, the ionic compound barium peroxide, BaO_2, is formed. Which dot-and-cross diagram represents the electronic structure of the peroxide anion in BaO_2?

Key: ○ electron from first oxygen atom × electron from second oxygen atom ● electron from barium atom

A 

B 

C 

D 

5 0.50 mol of I_2(g) and 0.50 mol of Br_2(g) are placed in a closed flask. The following equilibrium is established.

$$I_2(g) + Br_2(g) \rightleftharpoons 2IBr(g)$$

The equilibrium mixture contains 0.80 mol of IBr (g). What is the value of K_c?

A 0.64

B 1.3

C 2.6

D 64

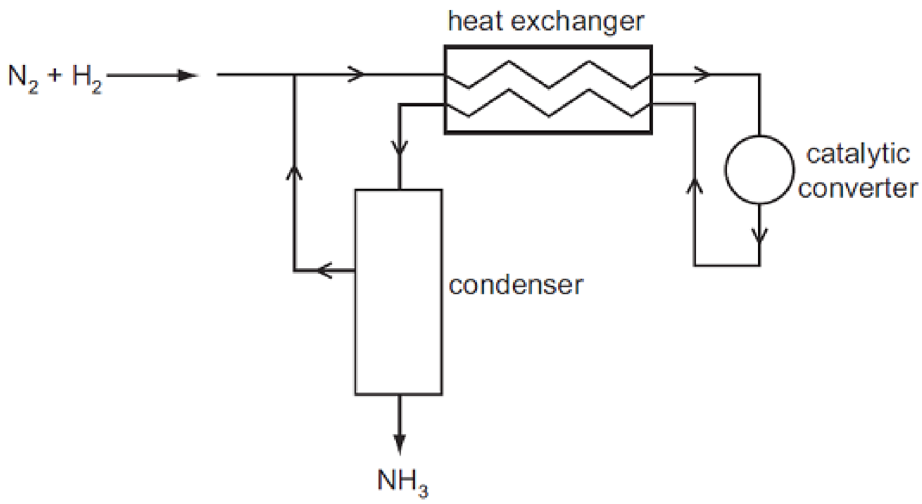
6 Skin cancer can be treated using a radioactive isotope of phosphorus, ^{32}P.

A compound containing the phosphide ion $^{32}P^{3-}$, wrapped in a plastic sheet, is strapped to the affected area.

What is the composition of the phosphide ion, $^{32}P^{3-}$?

	protons	neutrons	electrons
A	15	17	18
B	15	17	32
C	17	15	17
D	32	17	15

7 The diagram represents the Haber process for the manufacture of ammonia from nitrogen and hydrogen.



What is the purpose of the heat exchanger?

A to cool the incoming gas mixture to avoid overheating the catalyst

B to cool the reaction products and separate the NH_3 from unused N_2 and H_2

C to warm the incoming gas mixture and shift the equilibrium to give more NH_3

D to warm the incoming gas mixture and speed up the reaction

8 Which species behave as Brønsted-Lowry acids in the following reversible reaction?

$$H_2PO_4^-(aq) + CN^-(aq) \rightleftharpoons HCN(aq) + HPO_4^{2-}(aq)$$

A HCN and CN^-

B HCN and $H_2PO_4^-$

C HCN and HPO_4^{2-}

D $H_2PO_4^-$ and HPO_4^{2-}

9 The table shows the enthalpy change of neutralisation per mole of water formed, ΔH, for various acids and bases.

acid	**base**	**ΔH / kJ mol^{-1}**
sulfuric acid	sodium hydroxide	-57.0
P	sodium hydroxide	-54.0
sulfuric acid	**Q**	-52.0
R	potassium hydroxide	-57.0

What are **P**, **Q** and **R**?

	P	**Q**	**R**
A	sulfuric acid	sodium hydroxide	ethanoic acid
B	ethanoic acid	sodium hydroxide	phosphoric acid
C	sulfuric acid	ammonia	hydrochloric acid
D	ethanoic acid	ammonia	nitric acid

10 Gallium nitride, GaN, could revolutionise the design of electric light bulbs because only a small length used as a filament gives excellent light at low cost.

Gallium nitride is an ionic compound containing the Ga^{3+} ion.

What is the electron arrangement of the nitrogen ion in gallium nitride?

A $1s^2\ 2s^2$

B $1s^2\ 2s^2\ 2p^3$

C $1s^2\ 2s^2\ 2p^4$

D $1s^2\ 2s^2\ 2p^6$

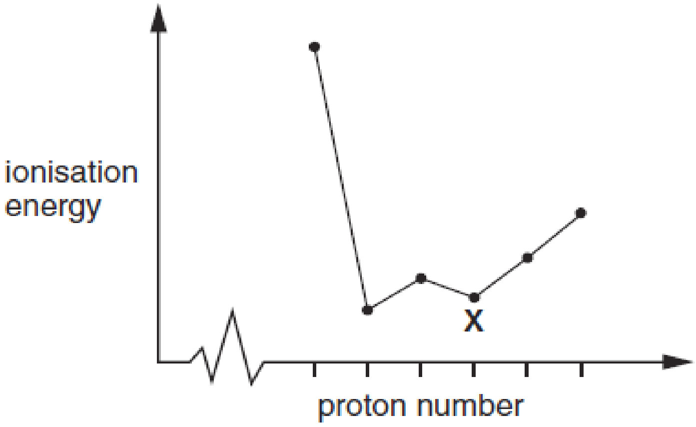
11 The table gives the successive ionisation energies for an element ***X***.

	1st	2nd	3rd	4th	5th	6th
ionisation energy/ kJ mol^{-1}	950	1800	2700	4800	6000	12300

What could be the formula of the chloride of ***X***?

A XCl **B** XCl_2 **C** XCl_3 **D** XCl_4

12 The sketch below shows the variation of first ionisation energy with proton number for six elements of consecutive proton numbers between 1 and 18 (H to Ar).



What is the identity of the element **X**?

A Mg **B** A*l* **C** Si **D** P

13 Which statement concerning only the elements in the third period, sodium to argon, is correct?

A The element that has four atoms in its molecule is sulfur.

B The element with the highest electrical conductivity is aluminium.

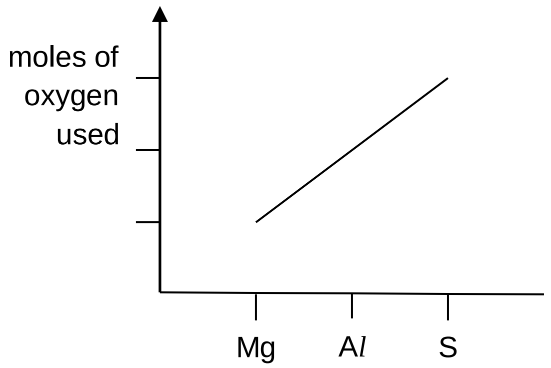
C The element with the highest melting point is aluminium.

D The element with the largest anion is chlorine.

14 One mole of magnesium, aluminium and sulfur are each completely burned in an excess of oxygen gas.

Which graph shows the moles of oxygen used in each case?

A



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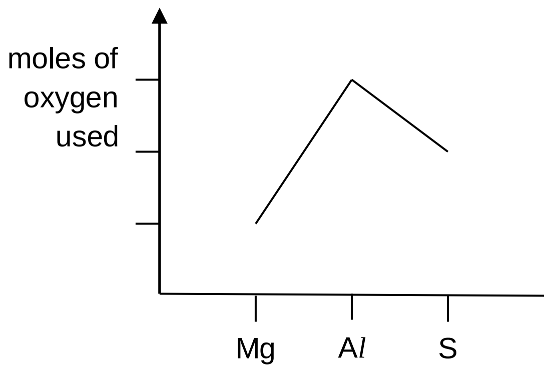

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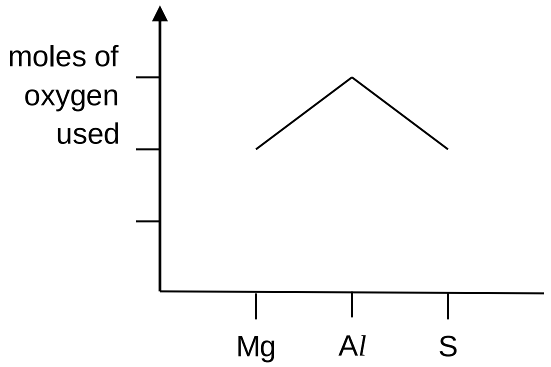

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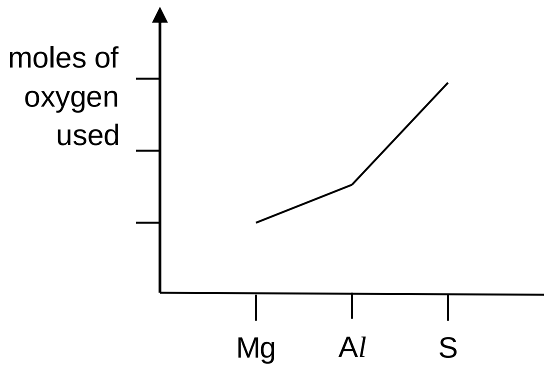
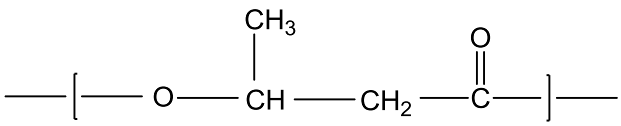


15 PHB (polyhydroxybutyric acid) is a natural polymer produced by a range of micro-organisms. It can also be manufactured from sugar. PHB is readily biodegradable.



PHB (polyhydroxybutyric acid)

What type of reaction will cause PHB to break down?

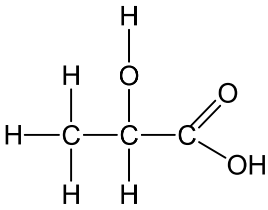
A addition

B hydrolysis

C reduction

D substitution

16 Lactic acid occurs naturally, for example in sour milk.
Its displayed formula is shown.



Which reaction occurs with lactic acid?

A It decolourises aqueous bromine rapidly.

B It is insoluble in water.

C It reduces Fehling's reagent.

D Two molecules react with each other in the presence of a strong acid.

17 What is formed when propanone is reacted with $NaBH_4$?

A propanal

B propan-1-ol

C propan-2-ol

D propane

18 Which of the following can exist in **both** polar and non-polar forms?

A CH_2Cl_2

B C_2HCl

C $C_2H_2Cl_2$

D C_2H_3Cl

19 The C_2N_2 molecule is linear.

What can be deduced from this about the numbers of σ and π bonds present in the molecule?

	σ	π
A	3	2
B	3	3
C	3	4
D	4	2

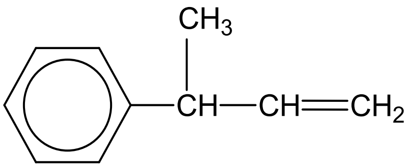
20 Compound ***X***

• has the molecular formula $C_{10}H_{14}O$;

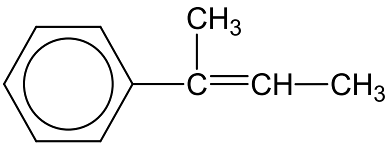
• is unreactive towards acidified $K_2Cr_2O_7$.

What is the structure of the compound formed by dehydration of ***X***?

A



B



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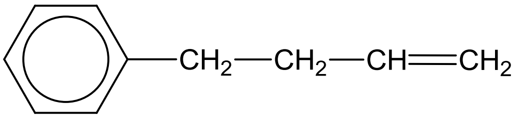
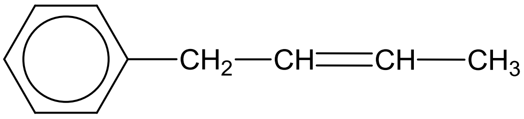


D



21 Dichlorodifluoromethane, CCl_2F_2, has been used in aerosol propellants and as a refrigerant.

Which statement helps to explain why dichlorodifluoromethane is chemically inert?

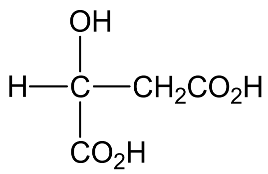
A The carbon-fluorine bond energy is large.

B The carbon-fluorine bond has a low polarity.

C Fluorine is highly electronegative.

D Fluorine compounds are non-flammable.

22 Malic acid occurs in apples.



malic acid

Which substance will react with all three –OH groups present in the malic acid molecule?

A ethanol in the presence of concentrated sulfuric acid

B potassium hydroxide

C sodium

D sodium carbonate

23 Which reagent could be used to distinguish between $CH_3CH(OH)CH_2CHO$ and $CH_3COCH_2CH_2OH$?

A Fehling's reagent

B dilute sulfuric acid

C 2,4–dinitrophenylhydrazine

D acidified potassium dichromate(VI)

24 A compound ***Y*** has all of the properties below.

- It is a liquid at 25°C.
- It mixes completely with water.
- It reacts with aqueous sodium hydroxide.

What could ***Y*** be?

A ethanoic acid

B ethanol

C ethane

D ethyl ethanoate

25 Compound **P**, $C_4H_{10}O_2$, is an alcohol containing two –OH groups per molecule. It can be oxidised in stages to compound **Q**, $C_4H_8O_2$, and then to compound **R**, $C_4H_6O_2$, which is a diketone.

Which of the following will react with alkaline aqueous iodine to give tri-iodomethane?

A **P, Q** and **R**

B **P** and **R** only

C **Q** and **R** only

D **P** and **Q** only

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, 2 and 3 are correct	1 and 2 are correct	2 and 3 are correct	1 only is correct

No other combination of statements is used as a correct response.

26 On a scale in which the mass of a ^{12}C atom is 12, the relative molecular mass of a particular sample of chlorine is 72.

Which properties of the atoms in this sample are always the same?

1 radius

2 nucleon number

3 isotopic mass

27 Hydroxyapatite, $Ca_5(PO_4)_3OH$, is the main constituent of tooth enamel. In the presence of saliva, the following equilibria exist.

$$Ca_5(PO_4)_3OH(s) \rightleftharpoons 5Ca^{2+}(aq) + 3PO_4^{3-}(aq) + OH^-(aq)$$

$$HPO_4^{2-}(aq) \rightleftharpoons H^+(aq) + PO_4^{3-}(aq)$$

Which of the following statements help to explain why tooth enamel is dissolved more readily when saliva is acidic?

1 The hydroxide ions are neutralised by the acid.

2 The phosphate ion $PO_4^{3-}(aq)$ accepts $H^+(aq)$.

3 Calcium ions react with acids.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 are correct	2 and 3 are correct	1 only is correct

No other combination of statements is used as a correct response.

28 Water is added to anhydrous aluminium chloride to make a 0.1 mol dm^{-3} solution. Which observations are correct?

1 The reaction is endothermic.

2 The solution is acidic.

3 The solution contains the ion $[Al(H_2O)_6]^{3+}$.

29 The conversion of CrO_4^{2-}(aq) into $Cr_2O_7^{2-}$(aq) is represented by the following equation.

$$\underset{\text{yellow}}{2CrO_4^{2-}(aq)} + 2H^+(aq) \rightleftharpoons \underset{\text{orange}}{Cr_2O_7^{2-}(aq)} + H_2O(l)$$

Which statements are correct?

1 Addition of OH^-(aq) to the equilibrium mixture causes a change of colour.

2 The conversion of CrO_4^{2-}(aq) into $Cr_2O_7^{2-}$(aq) involves a change of oxidation state.

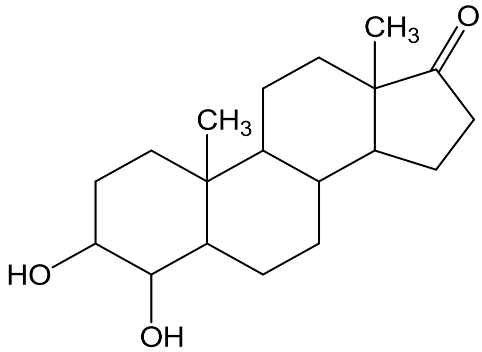
3 CrO_4^{2-}(aq) acts as a base.

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, 2 and 3 are correct	1 and 2 are correct	2 and 3 are correct	1 only is correct

No other combination of statements is used as a correct response.

30 The steroid shown is an intermediate compound obtained during the synthesis of Formestane which is used in the treatment of breast cancer.



Which statements about this compound are correct?

1 It reacts with hydrogen cyanide in an addition reaction.

2 It can be oxidised by warm acidified potassium dichromate(VI) to a carboxylic acid.

3 It will react with Fehling's solution.